Jack A. Logan

Theoretical and Computational Physicist

Mathematical Modeling | Numerical Algorithms | Scientific Computing

Salt Lake City, UT | Open to relocation | jalogan.github.io | LinkedIn | GitHub

Research Profile

Theoretical and computational physicist with a PhD in Physics and 9+ years of research experience developing mathematical theories, numerical algorithms, and scientific software. Research spans statistical mechanics, interacting-particle systems, molecular modeling, geometric constraints, and symmetry-based methods, with additional independent work in scientific machine learning. Led theoretical and computational development alongside doctoral advisor and coordinated validation with experimental and simulation collaborators. First/co-first author of five peer-reviewed research papers. Proficient in C++ and Python; experienced in connecting original derivations with computational tests and measurable outcomes.

Research Capabilities

Theory: Statistical mechanics | phase behavior | symmetry/group theory | tensor methods | asymptotic analysis | constrained systems

Simulation and algorithms: Molecular dynamics | Monte Carlo | constraint integration (RATTLE) | zero modes/rigidity matrices | numerical linear algebra | coarse-grained modeling

Scientific computing: Python | C++ | NumPy | SciPy | PyTorch | scikit-learn | LAMMPS | HPC/parallel computing | CuPy (CUDA GPU acceleration) | Git | Linux

Education

Stony Brook University, Stony Brook, NY — PhD Physics 2022; MA Physics 2016; BS Physics 2009

Long Island University, Brookville, NY — MS Applied Mathematics 2014, *Summa Cum Laude*

PhD Dissertation: *Geometric and Topological Aspects of Self-Assembly: from spheres and rods to designer particles*

Research Experience

Yale University, New Haven, CT 2022-2025

Postdoctoral Fellow, Mechanical Engineering & Materials Science

- Led development of six C++ coarse-grained protein model variants, investigating backbone geometry, angle/dihedral constraints, and sequence-dependent side-chain representations including modified Martini-inspired models.
- Selected and curated 2,531 experimentally characterized proteins, developed quantitative structural observables, and performed 15,186 model–protein simulation runs (2,531 for each of six models) to identify the physical constraints needed to reproduce experimental structure.
- Implemented CuPy GPU acceleration for structure-factor calculations and a custom Monte Carlo analysis estimating core packing fractions for irregularly shaped amino acid models.
- Led the scientific modeling effort and worked with student researchers; results published as first author *Physical Review E* Editors' Suggestion (2025).

Brookhaven National Laboratory, Upton, NY 2016-2022

Research Assistant, Center for Functional Nanomaterials

Doctoral research at BNL during Stony Brook University graduate study (overlapping dates).

- Developed original statistical-mechanical theories and computational algorithms for constrained sphere packings, particle interactions, and symmetry-adapted structural characterization, including topology-changing Monte Carlo methods and analytical interaction potentials.
- Led theoretical and computational analysis in multi-institution research collaborations, developing mathematical frameworks, structural classification algorithms, and quantitative methods to characterize and validate ordered and disordered nanoparticle structures against simulations and experiments.

Selected Peer-Reviewed Research

SymBOPs: Symmetry-Resolved Structural Order
Group theory | Tensor and spherical-harmonic representations | Structural algorithms | Experimental validation

- Developed a mathematical framework of symmetry-adapted tensor order parameters using group theory and spherical harmonics to characterize crystalline symmetry, coherent amorphous order, and local orientational structure.
- Developed bond-level classification and domain-detection algorithms to identify ordered regions, interfaces, and structural defects; applied and validated the framework using simulated and experimental nanoparticle data across three peer-reviewed studies.

Geometric and Topological Entropies of Sphere Packing
Statistical mechanics | Rigidity/zero modes | New Monte Carlo sampling method

- Developed a statistical-mechanical framework separating geometric and topological contributions to the entropy of sphere packings, investigating how mechanical constraints and contact-network topology govern disordered configurations.
- Designed an isostaticity-preserving Monte Carlo algorithm that explores distinct packing topologies by breaking contacts, following zero-mode displacements, and forming new contacts; implemented column-pivoted QR for numerically robust solutions of constrained linear systems.

Analytical Interaction Potential for van der Waals Nanorods
Analytical physics | Asymptotic limits | Efficient numerical models

- Derived analytical approximations for interaction potentials between finite nanorods at arbitrary relative positions and orientations; numerically validated their accuracy across geometric regimes, reducing computationally expensive multidimensional integrations while preserving near- and far-field behavior.

Hybrid Isotropic/Patchy-Particle Self-Assembly
Physical theory | Particle morphology | Collaborative modeling

- Led theoretical and computational analysis of a collaborative self-assembly study, developing symmetry-based structural characterization and scientific visualizations to characterize how particle and interaction parameters influence the formation of crystalline and coherent amorphous morphologies in collaborator-generated LAMMPS simulations.

Structural Properties of Folded Proteins
Co-first-author research | C++ molecular dynamics | 15,186 model–protein simulations

- Led development and computational validation of six coarse-grained protein models, performing 15,186 molecular-dynamics simulations across 2,531 proteins and implementing GPU-accelerated analysis of structural observables.
- Systematically investigated how backbone stereochemistry, side-chain geometry, and molecular constraints influence folded-protein structure, identifying model features required to reproduce key structural properties of experimentally determined proteins.

Independent Computational Research

Jamming on Curved Manifolds
Constrained dynamics | C++ simulation | Differential geometry

- Developed a custom C++ molecular-dynamics engine using RATTLE constraints for three-dimensional spheres tangent to planar, spherical, cylindrical, and toroidal substrates; implemented surface-aware interactions, jamming procedures, and custom surface-meshing algorithms to investigate geometrically constrained particle systems.

Physics-Informed Neural-Network Force Field
PyTorch | LAMMPS | Learning dynamical laws | Model validation

- Developed a neural-network surrogate for Lennard–Jones pair interactions that preserves rotational equivariance and Newton's third law; integrated the learned forces into molecular dynamics and evaluated force accuracy, trajectory divergence, and ensemble-level structural observables, including radial distribution functions.

Mechanistic Analysis of Emergent D_3 Symmetry in a Transformer
PyTorch | Group representations | Activation interventions

- Trained a one-layer Transformer on operations of the dihedral group D_3 and designed activation-reinjection experiments to investigate learned operator composition; examined internal representations and composition failures using sparse autoencoders, hidden-state similarities, and principal component analysis.

Participation Ensembles for Interpretable Neural Representations
Information theory | Representation analysis | Work in progress

- Formulated event–component probability distributions and conditional-entropy diagnostics for superposition, polysemanticity, and reuse of neural representations; proposed targeted tests of these measures.

Selected Publications

- **Logan***, Sumner*, Grigas, Shattuck, and O'Hern, "Effect of stereochemical constraints on the structural properties of folded proteins," *Physical Review E* **112**, 054405 (2025). **Editors' Suggestion; equal first authors**. doi:10.1103/9wf9-ywhw.
- **Logan**, Michelson, Pattammattel, Yan, Gang, and Tkachenko, "Symmetry-specific characterization of bond orientation order in DNA-assembled nanoparticle lattices," *Journal of Chemical Physics* **159**, 154905 (2023). doi:10.1063/5.0168604.
- **Logan**, Mushnoori, Dutt, and Tkachenko, "Symmetry-specific orientational order parameters for complex structures," *Journal of Chemical Physics* **156**, 054108 (2022). doi:10.1063/5.0076915.
- **Logan** and Tkachenko, "Geometric and topological entropies of sphere packing," *Physical Review E* **105**, 014117 (2022). doi:10.1103/PhysRevE.105.014117.
- Mushnoori, **Logan**, Tkachenko, and Dutt, "Controlling morphology in hybrid isotropic/patchy particle assemblies," *Journal of Chemical Physics* **156**, 024501 (2022). **Editors' Pick**. doi:10.1063/5.0076914.
- **Logan** and Tkachenko, "Compact interaction potential for van der Waals nanorods," *Physical Review E* **98**, 032609 (2018). doi:10.1103/PhysRevE.98.032609.

Additional Publications

- Grigas, Liu, **Logan**, Shattuck, and O'Hern, "Protein Folding as a Jamming Transition," *PRX Life* **3**, 013018 (2025). doi:10.1103/PRXLife.3.013018.
- Gilbert Corder et al. (including **Logan**), "Near-field spectroscopic investigation of dual-band heavy fermion metamaterials," *Nature Communications* **8**, 2262 (2017). doi:10.1038/s41467-017-02378-3.

Selected Research Leadership & Awards

- Led modeling and theoretical analysis in multi-institution research collaborations involving experimental scientists and simulation groups.
- Team Leader, Protein Folding Simulation Project, Yale University; mentored student summer researchers.
- GSNP Postdoctoral Speaker Award Finalist, APS March Meeting; Jonathan Kaufman Student Excellence Prize.
- Yale postdoctoral and Physical and Engineering Biology (PEB) research travel awards.